

Boundary-spanning search and firms' green innovation: The moderating role of resource orchestration capability

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Abstract

Building on resource-based theory and resource orchestration theory, we investigate the impact of two characteristics of boundary-spanning search, search breadth and search depth, on firms' exploitative and exploratory green innovations. We also examine the moderating role of resource orchestration capability. The results of data analysis from 186 manufacturing firms in China show that both search breadth and search depth have inverted U-shaped relationships with exploitative and exploratory green innovations. Furthermore, resource orchestration capability is found to moderate the inverted U-shaped relationship between boundary-spanning search and green innovation. Specifically, with high resource orchestration capability, the inverted U-shaped relationships of search breadth with exploitative and exploratory green innovations are flattened, whereas the relationships of search depth with exploitative and exploratory green innovations are almost linear. Our research contributes to the literature on green innovation by uncovering the complex effects of boundary-spanning search on exploitative and exploratory green innovations.

KEYWORDS

boundary-spanning search, exploitative green innovation, exploratory green innovation, resource orchestration capability, sustainable development

1 | INTRODUCTION

Green innovation has increasingly attracted firms' and scholars' attention. Many firms begin to regard green innovation as an effective strategy to obtain sustainable competitive advantage (Du, Zhang, & Feng, 2018; Pickett-Baker & Ozaki, 2008). Comparing with traditional innovations, green innovation is a new innovation model to deal with energy saving, pollution preventing, waste recycling, green product design, and corporate environmental management (Tang, Walsh, Lerner, Fitza, & Li, 2018; Tsai & Liao, 2017). The aim of green innovation is to produce novel or improved products and processes that diminish negative environmental impacts and eventually realize the harmonious co-development of environment, economy, and society (Al Mamun, Mohamad, Yaacob, & Mohiuddin, 2018). According to ambidexterity theory, green innovation can be divided into exploitative green innovation and exploratory green innovation, and the two types of innovation are

conducted at the same time within a firm (O'Reilly & Tushman, 2008). Explorative green innovation involves activities for discovering and exploiting existing environmental knowledge and experience to improve current green products, processes, and services, whereas exploratory green innovation engages in searching and exploring new environmental knowledge and experience to develop new green products, processes, and services. Therefore, green innovation can not only help firms to develop environmental products and processes but also play an important role in undertaking social responsibilities and hindering competitors' imitation (Chang, 2011), and thus acquiring first-mover advantages (Chen & Chang, 2013).

Green innovation is a new type of innovation by which firms implement new practice, processes, and products to prevent environmental damage and facilitate waste recycling (Chen, Cheng, & Dai, 2017). Current research suggests that factors, such as institutional pressures (Lin & Ho, 2016), environmental regulation (Hojnik & Ruzzier, 2016;

You, Zhang, & Yuan, 2019; Zhang, Wang, Xue, & Yang, 2018), stakeholder pressure (Kawai, Strange, & Zucchella, 2018), external knowledge resources (Arfi, Hikkerova, & Sahut, 2018; Laursen & Salter, 2006), big data and environmental ethics (El-Kassar & Singh, 2019), and market demands (Mondéjar-Jiménez, Segarra-Oña, Peiró-Signes, Payá-Martínez, & Sáez-Martínez, 2015), influence firms' green innovation. As a complex and costly technological frontier (De Marchi, 2012; Liao, 2018), green innovation requires more diversified knowledge resources than traditional innovations (Cainelli, De Marchi, & Grandinetti, 2015; Martínez-Ros & Kunapatarawong, 2019; Pacheco, Alves, & Liboni, 2018), especially complex technological knowledge and skills for eliminating environmental pollution (Liao & Tsai, 2019). However, due to insufficient internal knowledge resources, firms have to search for external knowledge to conduct green innovation (Du et al., 2018). According to resource-based theory, valuable, rare, and inimitable resources determine the firm's sustainable competitive advantage (Barney, 1991), and the firm needs to explore, utilize, and integrate different special knowledge from external R&D activities and internal resources (Grant, 1996). As suggested by Zimmermann, Ferreira, and Moreira (2016), a firm needs to obtain knowledge resources from external partners to enhance innovation. Ardito, Petruzzelli, Pascucci, and Peruffo (2019) and Liao and Long (2019) also highlighted the importance of cooperation with external partners, which enable firms to acquire external knowledge and develop green innovation. Moreover, external knowledge obtained from stakeholders offers the firm more opportunities to recognize customers green needs and facilitates the implementation of green innovation (Albort-Morant, Leal-Millán, & Cepeda-Carrión, 2016; Dangelico, Pujari, & Pontrandolfo, 2011). Thus, external knowledge is particularly important for green innovation.

Boundary-spanning search, as an effective way to acquire external knowledge, is the process through which firms span organizational boundaries to search and identify new knowledge (Classen, van Gils, Bammens, & Carree, 2012; Ren, Eisingerich, & Tsai, 2015; Rosenkopf & Nerkar, 2001). It enables firms to overcome organizational rigidity (Leonard-Barton, 1992) and capability trap (Levitt & March, 1988). At the same time, boundary-spanning search is likely to facilitate firms to better fit with the changing environment, helps firms obtain heterogeneous knowledge resources, complements firms' internal resources, increases firms' knowledge diversity, and eventually creates more opportunities for firms to conduct green innovation (De Marchi & Grandinetti, 2013; Zhang & Hu, 2017; Zhao, Feng, & Shi, 2018). Hence, boundary-spanning search has become a key driver of firms' green innovation. Boundary-spanning search has two characteristics: search breadth and search depth (Katila & Ahuja, 2002; Laursen & Salter, 2006). Search breadth shows the wide range of external search, whereas search depth describes the depth of external search. Both of them provide opportunities for firms to gain new knowledge, overcome resources limitation, and conduct innovation (Laursen & Salter, 2006). Existing research has found the relationship between boundary-spanning search and innovation, yet the results are still inconsistent. Some scholars argue that boundary-spanning search fosters innovation

(Leiponen & Helfat, 2010; Ruiz-Pava & Forero-Pineda, 2018; Zhang, Li, & Zheng, 2017), whereas others suggest boundary-spanning search may have inverted U-shaped relationship with innovation (Ardito & Petruzzelli, 2017; Laursen & Salter, 2006; Zhang & Hu, 2017). In addition, boundary-spanning search may hamper innovation (Terjesen & Patel, 2017) or have no significant effect on innovation (Ferreras-Méndez, Fernández-Mesa, & Alegre, 2016). Hence, it is necessary to clarify the contradictory relationship between boundary-spanning search and innovation. Although boundary-spanning search is found to be an important antecedent of innovation in the context of environmental sustainability, there is little research on the impact of boundary-spanning search on green innovation. Although Arfi et al. (2018) and Martínez-Ros and Kunapatarawong (2019) discussed the impact of external knowledge resources on green innovation, Arfi et al. (2018) emphasized the relationship between internal and external knowledge sharing with green innovation by using a case study, and Martínez-Ros and Kunapatarawong (2019), who conducted an empirical study, only addressed the impact of search breadth on green innovation and did not consider the dimensions of green innovation. Moreover, the relationship between boundary-spanning search and innovation may be different in the environmental sustainability context. Therefore, it is necessary to further explore how boundary-spanning search influences green innovation and examine whether there are double-edged sword effect between the two search characteristics and exploitative and exploratory green innovations.

Despite external knowledge resources are important, firms need resource orchestration capability to effectively assess and apply the knowledge (Sirmon, Hitt, & Ireland, 2007). Resource orchestration, a promising area of investigation to understand how firms derive benefits from their resources, has received great attention in recent years. Previous studies show that orchestration of resources is important for adapting to changing market conditions and for improving innovation (Chadwick, Super, & Kwon, 2015; Sirmon et al., 2007; Wales, Patel, Parida, & Kreiser, 2013; Wong, Wong, & Boon-itt, 2018). For example, from the resource orchestration viewpoint, Sirmon et al. (2007) emphasized the role of firms' internal capabilities in structuring, bundling, and leveraging the resources portfolio. Wales et al. (2013), Teece (2014), and Gong, Jia, Brown, and Koh (2018) indicated that resources should be orchestrated. A study by Chadwick et al. (2015) also highlighted the role of firms in orchestrating resources. Resource orchestration is critical to decrease internal conflict and to increase resources complementarities in the firm (Teece, 2007), which is a kind of dynamic capability for helping firms to transform new knowledge to facilitate green innovation (Albort-Morant, Leal-Rodríguez, & De Marchi, 2018; Hashim, Bock, & Cooper, 2015). Building on the research of Sirmon et al. (2007); Sirmon, Hitt, Ireland, and Gilbert (2011); and Queiroz, Tallon, Sharma, and Coltman (2018), we regard resource orchestration capability as the firms' internal capability that combine, configurate, and deploy knowledge resources to develop exploitative and exploratory green innovations. We also suggest that resource orchestration capability moderates the link between boundary-

spanning search and green innovation. Unfortunately, when studying the innovation process, prior research mostly focused on firms' absorptive capability (Flor, Cooper, & Oltra, 2018; Pacheco et al., 2018; Sheng & Chien, 2016), combinative capability (Sheng, 2017), and social media capability (Benitez, Castillo, Llorens, & Braojos, 2018) but neglected resource orchestration capability. As an important capability to facilitate green innovation, resource orchestration capability deserves more research, and it may help to explain the contradictory results between boundary-spanning search and innovation. Thus, we include resource orchestration capability in this study and intend to uncover the influencing mechanism of resource orchestration capability on the relationship between boundary-spanning search and exploitative and exploratory green innovations.

To advance research on green innovation, we examine the impact of search breadth and search depth on exploitative and exploratory green innovations and analyze the moderator role of resource orchestration capability. This study makes two contributions. First, we found that the two characteristics of boundary-spanning search have inverted U-shaped relationships with both exploitative and exploratory green innovations. Second, we discovered that resource orchestration capability played a moderating role between boundary-spanning search and green innovation. Specifically, with a high level of resource orchestration capability, the inverted U-shaped relationships of search breadth with exploitative and exploratory green innovations are flattened, whereas the relationships of search depth with exploitative and exploratory green innovations are almost linear.

The rest parts are structured as follows. First, we describe the theoretical background of our research and develop hypotheses. Then, we report data collection and variables measurement. Third, we present the results of hypotheses testing. Last, we discuss the theoretical and practical implications and research limitations.

2 | THEORETICAL BACKGROUND AND HYPOTHESES

Drawing on resource-based theory and resource orchestration theory, we suggest that boundary-spanning search plays an important role in green innovation and may have impact on both types of green innovation, namely, exploitative green innovation and exploratory green innovations. We also examine the moderating role of resource orchestration capability. Figure 1 shows our theoretical model.

2.1 | Boundary-spanning search and green Innovation

2.1.1 | Search breadth and green innovation

Search breadth is the width that firms span organization boundary to access different fields of knowledge. Regarding the link between search breadth and innovation, some research suggests that search breadth positively affects innovation (Leiponen & Helfat, 2010; Martínez-Ros & Kunapatarawong, 2019; Zhang et al., 2017). As previously proposed, search in a wide range may increase a firm's knowledge diversity and enrich its knowledge base, provide the firm with more options to solve problems (March, 1991), and obtain new insights to facilitate exploratory innovation (Flor et al., 2018). However, some researchers suggest that knowledge breadth negatively impacts on innovation (Terjesen & Patel, 2017). On the one hand, firms with a wider search may require large combination and maintenance efforts (Cruz-González, López-Sáez, Navas-López, & Delgado-Verde, 2015; Martínez-Ros & Kunapatarawong, 2019) to combine external new knowledge and/or maintain relationships with partners (Henderson & Clark, 1990; Sisodiya, Johnson, & Grégoire, 2013), which might limit external knowledge transferred into green innovation processes and hamper green innovation. On the other hand, wider search may consume excessive organizational and managerial energy (Faems, De Visser, Andries, & Van Looy, 2010; Martínez-Ros & Kunapatarawong, 2019). Firms then have less energy to utilize external knowledge and result in decreasing creativity. Moreover, compared with similar knowledge, external diversified knowledge is difficult to access and integrate (Cruz-González, López-Sáez, Emilio Navas-López, & Delgado-Verde, 2014). Thus, knowledge coming from a wide range will incur many constraints and limitations for the innovation (Zhang & Hu, 2017).

We postulate that search breadth may have double-edged sword effect on green innovation. On the one hand, when the degree of search breadth is moderate, exploitative and exploratory green innovations are more likely to take place. The moderate degree of search breadth provides the firm more opportunities to acquire diverse knowledge; the broader knowledge provides firms chances to flexibly adapt to environmental change (Cruz-González et al., 2014; Huang & Li, 2018) and finally promotes green innovation. On the other hand, when the degree of a firm's search breadth is too low or too high, green innovation will be reduced. Too narrow search may result in the situation that the acquired knowledge cannot satisfy the need of

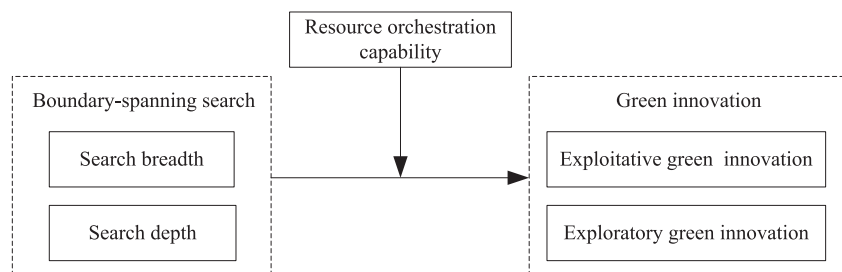


FIGURE 1 Theoretical model

green innovation, whereas too wide search crossing industries not only involves additional search efforts (Cruz-González et al., 2015) but also increases the difficulty of knowledge combination, and thus, leads to innovation reduction (Patel & Van der Have, 2010; Zhang & Hu, 2017). As such, we suggest that only when the search breadth is at a moderate level, the exploitative and exploratory green innovations can be facilitated. Therefore, we hypothesize the following:

Hypothesis 1a. *Search breadth has an inverted U-shaped relationship with exploitative green innovation.*

Hypothesis 1b. *Search breadth has an inverted U-shaped relationship with exploratory green innovation.*

2.1.2 | Search depth and green innovation

In contrast to search breadth, search depth is the depth that firms span organization boundary to access external knowledge. Search depth may have a positive effect on innovation (Katila & Ahuja, 2002; Terjesen & Patel, 2017). First, the similar external knowledge can not only reduce the likelihood of errors and failures (Laursen & Salter, 2006) but also assist the development of organizational routines to enhance the reliability of search (Levinthal & March, 1981). Second, experience accumulated in the past can increase the confidence of deeper search. The external knowledge can be acquired more quickly and effectively when the knowledge is familiar to the firm and suits for the firm's need, which in turn facilitates innovation. Third, the similar external knowledge enables the firm to deeply understand the knowledge and promotes the firm's capability to identify knowledge value, discovers knowledge relationship, and transfers internal and external knowledge into new ideas. However, search depth may also negatively impact on innovation if it is excessive (Laursen & Salter, 2006; Shi, Zhang, & Zheng, 2019). First, deeper search hinders the development of technology. When the search depth surpasses an optimal level, the search costs will overshadow the diminishing value of the new knowledge. Eventually, the effort of knowledge search is far higher than the benefits (Dosi, 1988). Second, organization will become rigid when search depth exceeds a certain level. For example, in order to maintain status quo, employees may hide the problem caused by existing knowledge, and the firm can become more and more rigid by keeping searching for similar knowledge (Cruz-González et al., 2014; Leonard-Barton, 1992). As Laursen and Salter (2006) found, excessively deep knowledge can be a disadvantage to innovation for firms.

We infer that search depth may have a curvilinear effect on green innovation. Specifically, when the degree of a firm's search depth is moderate, the knowledge about the same industry can be accessed, the similar knowledge enables professional development (Rosenkopf & Nerkar, 2001) and enhances the firm to economize green activities (Huang & Li, 2018). As a result, it facilitates firms to pursue both exploitative and exploratory green innovations. Contrarily, when search depth is too low or too high, green innovation will decline. Too shallow search may result in insufficient knowledge, which may hinder exploitative and exploratory green innovations. Too deep

search in the same field could lead to knowledge redundancy, which is potentially disadvantageous to green innovation. In sum, we propose that only search depth in a moderate level can enhance the exploitative and exploratory green innovations. Therefore, we hypothesize the following:

Hypothesis 2a. *Search depth has an inverted U-shaped relationship with exploitative green innovation.*

Hypothesis 2b. *Search depth has an inverted U-shaped relationship with exploratory green innovation.*

2.2 | The moderating role of resource orchestration capability

The capability of utilizing external knowledge determines the level of innovation (El-Kassar & Singh, 2019; Ferreras-Méndez et al., 2016). According to resource-based theory and resource orchestration theory, innovation and sustainable competitive advantage depend on the firm's internal capability (Barney, 1991; Chadwick et al., 2015; Chang, 2018; Sirmon et al., 2007). Hence, resource orchestration capability may play a role in the curvilinear relationship between boundary-spanning search and green innovation. For example, Flor et al. (2018) found that absorptive capacity has a positive impact on the relationship between external knowledge search breadth and depth and radical innovation. Albort-Morant et al. (2018) proposed that the firm's capability to integrate external knowledge is vital to green innovation. Hence, some researchers have called for more attention to the firm's ability to transform resources in green innovation (Huang & Li, 2018). Knowledge acquired by boundary-spanning search does not provide direct benefit for firms, and the firm's current knowledge and capability determine how much external knowledge can be utilized by the firm (Fabrizio, 2009). As Sheng (2017) found, the capability of orchestrating knowledge resources is vital for firms' exploitative and exploratory innovations. Similarly, Huang, Yang, and Wong (2016) suggested that green capability plays an important role for firms to adopt green product innovation. Hadjimanolis (2000) proposed that internal capabilities are vital for firms to achieve ambidextrous innovation. Hence, firms that engage in exploitative and exploratory green innovations rely not only on boundary-spanning search but also on the capability of integrating and transforming knowledge resources (Russo & Fouts, 1997). We therefore propose that resource orchestration capability plays a moderation role in the inverted U-shaped relationships between boundary-spanning search and exploitative and exploratory green innovations.

Specifically, we argue that resource orchestration capability may flatten the inverted U-shaped relationships between search breadth and exploitative and exploratory green innovations. When resource orchestration capability is high, whether conducting narrow search or wide search, firms will be able to better identify useful knowledge from external heterogeneous knowledge sources for innovation (Luo, Lui, & Kim, 2017), and can better structure, bundle and leverage the valuable external knowledge to generate novel ideas and thoughts

(Wales et al., 2013), thus facilitating green innovation. This is because high resource orchestration capability increases the positive effect of narrow search and reduces the negative effect of wider search on exploitative and exploratory green innovations. On the contrary, when resource orchestration capability is low, firms with narrow search may fail to explore and recognize useful knowledge (Zhou & Li, 2012) and then have fewer chances to assimilate and utilize external knowledge to create novel ideas (Inkpen & Pien, 2006; Lane, Koka, & Pathak, 2006), which reduce their green innovation. Moreover, due to low resource orchestration capability, firms with wider search may encounter many barriers to integrate and absorb the diverse external knowledge and may be overloaded in identifying valuable knowledge from too much knowledge (Ferreras-Méndez et al., 2016). Then, their green innovation is hampered by the difficulty of orchestrating wider range of external knowledge. Hence, low resource orchestration capability reduces the positive effect of narrow search and increases the negative effect of wider search on exploitative and exploratory green innovations. Taken together, the impacts of search breadth on exploitative and exploratory green innovations are more positive and are less negative when firms' resource orchestration capability is high than low. As such, we propose the following hypotheses:

Hypothesis 3a. *Resource orchestration capability moderates and flattens the inverted U-shaped relationship between search breadth and exploitative green innovation.*

Hypothesis 3b. *Resource orchestration capability moderates and flattens the inverted U-shaped relationship between search breadth and exploratory green innovation.*

We also suggest that resource orchestration capability may flatten the inverted U-shaped relationships between search depth and exploitative and exploratory green innovations. If resource orchestration capability is high, on the one hand, when conducting shallow search, firms can better assemble and extract the limited external knowledge from shallow search to meet their knowledge needs (Ritala & Hurmelinna-Laukkanen, 2013; Wales et al., 2013) and come up with new thoughts to accelerate green innovation. On the other hand, when conducting deep search, firms with high orchestration capability are able to understand and allocate the large amount of similar in-depth knowledge to satisfy the requirements of green innovation activities (Segarra-Ciprés & Bou-Lluis, 2018), thereby facilitating green innovation. Thus, the positive impacts of shallow search on green innovation will be strengthened, and the negative effect of deep search on green innovation will be mitigated or even disappear under high resource orchestration capability. By contrast, when resource orchestration capability is low, firms with shallow search may find it difficult to take advantage of the superficial external knowledge (Inkpen & Pien, 2006), and resulting in lower green innovation. Moreover, due to low resource orchestration capability, deep search may lead to a narrow knowledge base (Petruzzelli & Rotolo, 2015), and thus, firms are limited to single-dimensional external knowledge, which hamper green innovation. Furthermore, facing low resource

orchestration capability, firms with deep search cannot structure and bundle knowledge effectively (Wales et al., 2013). Additional efforts and time on orchestrating the similar but increasingly complex knowledge tend to increase their costs of knowledge assimilation, which can demotivate them from green innovations. Consequently, the impacts of shallow search on green innovation will be decreased, and the effects of deeper search on green innovation will be increased under low resource orchestration capability. Taken together, the impacts of search depth on exploitative and exploratory green innovations are more positive and are less negative when firms' resource orchestration capability is high than low. As such, we propose the following hypotheses:

Hypothesis 3c. *Resource orchestration capability moderates and flattens the inverted U-shaped relationship between search depth and exploitative green innovation.*

Hypothesis 3d. *Resource orchestration capability moderates and flattens the inverted U-shaped relationship between search depth and exploratory green innovation.*

3 | METHOD

3.1 | Data collection

The manufacturing industry plays a leading role in China's economy development, and carbon dioxide emissions by this industry accounts for more than 50% of the total (Zhao, Chen, Hayat, Alsaedi, & Ahmad, 2014). For China's manufacturing industry, it is urgent to conduct green innovation and upgrade from extensive growth to green development. We therefore selected the manufacturing industry as our research context. At first, we designed questionnaires in English and then translated them into Chinese by two scholars who have research experience abroad. We then selected 300 manufacturing firms in Shenzhen, Shanghai, Beijing, and Xi'an. Shenzhen, located in Southern China, is the pioneer city of reform and opening up of China and has lots of industrial firms. Shanghai, located in Eastern China, is an international economic center. Beijing, located in Northern China, is the capital of China and has taken measures to facilitate green activities. Xi'an is the most important industrial base in Western China. These four large cities represent the industry development of China in different geographical locations, which provide a holistic view of China's green innovation development. We contacted the top or middle managers of each firm and asked for their help to fill out the questionnaires. We administered the questionnaire by face-to-face interviews and postal mail. A reminder call was made about 2 weeks after the postal mail to improve response rate. In total, we obtained 198 questionnaires. After deleting incomplete responses, we received 186 usable questionnaires, resulting in an effective response rate of 62%.

Among the 186 respondent firms, 18 (9.68%) are from the textile industry, 34 (18.28%) the papermaking industry, 45 (24.19%) the chemical industry, 56 (30.11%) the equipment industry, and 33 (17.74%) other manufacturing industries. In terms of firm age, 15 (8.06%) are less

than 5 years, 26 (13.98%) from 5 to 10 years, 35 (18.82%) from 11 to 15 years, 64 (34.41%) from 16 to 20 years, and 46 (24.73%) more than 20 years. Regarding firm scale, 16 (8.60%) have fewer than 500 employees, 42 (22.58%) from 500 to 1,000 employees, 43 (23.12%) 1,001 to 2,000 employees, 67 (36.02%) 2,001 to 3,000 employees, and 18 (9.68%) more than 3,000 employees.

To test non-response bias, we applied the *t*-test to compute the mean differences in all variables between earlier and later respondents. The results yielded no significant difference, indicating non-response bias is not a serious problem in our research. Because the data for our research were received from single respondents, there may be common method bias in our research. We took three measures to address this concern. First, we randomized the order of items in the questionnaire and assured the respondents that all answers were anonymous to improve honest opinions. Second, based on the recommendation of Podsakoff, MacKenzie, Lee, and Podsakoff (2003), we conducted Harman's single-factor test to assess common method bias. A five-factor solution with eigenvalues bigger than one was obtained by putting all the factors into a principal component analysis. The total variance was 83.56%, and the first factor accounted for only 17.41% of the variance. Third, we examined the common method bias by applying confirmatory factor analysis. The results displayed that the fit of one-factor model with $\chi^2 = 2,251.112$, RMSEA = 0.280, TLI = 0.265, and CFI = 0.347 were worse than that of the five-factor model. Common method bias hence does not seem to be a serious threat in our research.

3.2 | Measures

We designed the survey questionnaire based on the existing literature and refined each items through in-depth interviews with five managers and scholars. Each item was rated by using a 5-point Likert scale.

For boundary-spanning search, we used the items of Katila and Ahuja (2002) and Zhang and Hu (2017). A 4-item scale was developed to measure search breadth, which includes the following: our firm owns many channels to search knowledge about a new field; our firm can search new field knowledge about R&D, production, and marketing; our firm can search new field knowledge about technology and management; and our firm can acquire more new knowledge in search. To measure search depth, four items were also designed: our firm owns many channels to search knowledge about an existing field; our firm can conduct in-depth search and obtain knowledge of an existing field about R&D, production, and marketing; our firm can conduct in-depth search and obtain knowledge of an existing field about technology and management; and our firm can apply the obtained knowledge of an existing field.

To reflect resource orchestration capability, we adapted the items from Sirmon et al. (2011) and Queiroz et al. (2018). Three items were used: our firm has capability to absorb all kinds of knowledge resources; our firm has capability to integrate all kinds of knowledge resources; and our firm has capability to utilize all kinds of knowledge resources.

To measure exploitative green innovation and exploratory green innovation, we used the items from Jansen, Van Den Bosch, and Volberda (2006) and He and Wong (2004). Exploitative green innovation was measured by the following statements: our firm improves current green products, processes, and services; our firm adjusts current green products, processes, and services; our firm extends current green market; and our firm strengthens current green technology. Meanwhile, exploratory green innovation was assessed with four items: our firm adopts new green products, processes, and services; our firm exploits new green products, processes, and services; our firm discovers new green market; and our firm enters new green technology.

Three control variables were included: industry type, firm age, and firm scale. Industry type was measured by five dummy variables: textile,

TABLE 1 Reliability and construct validity

Variable	Items	Factor loading	Cronbach's α	Composite reliability
Search breadth (SB)	SB1	0.860	0.920	0.925
	SB2	0.828		
	SB3	0.877		
	SB4	0.911		
Search depth (SD)	SD1	0.900	0.913	0.923
	SD2	0.826		
	SD3	0.864		
	SD4	0.873		
Resource orchestration capability (ROC)	ROC1	0.926	0.948	0.935
	ROC2	0.911		
	ROC3	0.890		
Exploitative green innovation (EIGI)	EIGI1	0.856	0.929	0.920
	EIGI2	0.855		
	EIGI3	0.870		
	EIGI4	0.867		
Exploratory green innovation (ERGI)	ERGI1	0.809	0.930	0.912
	ERGI2	0.837		
	ERGI3	0.865		
	ERGI4	0.884		

papermaking, chemical, equipment, and other, coded as Industries 1 to 4, with other type manufacturing industries as the reference group. Firm age was expressed in years at the time of data collection, ranging from less than 5 to over 20 years. Firm scale was based on the number of employees ranging from less than 500 to over 3,000.

3.3 | Reliability and validity

First, we tested the reliability using Cronbach's α and composite reliability. As displayed in Table 1, the values of Cronbach's α and composite reliability for all five variables were larger than the threshold value of 0.70, indicating acceptable reliability. Second, we examined the construct validity by conducting exploratory factor analysis with varimax rotation. The results showed that five factors with eigenvalues above 1 were extracted, and the factor loadings were greater than 0.50 (Nunnally, 1978), demonstrating acceptable construct validity.

Third, we assessed the convergent validity by performing confirmatory factor analysis. The results yielded a reasonable fit with $\chi^2 = 382.246$, RMSEA = 0.096, TLI = 0.910, and CFI = 0.925. The five-factor model hence was acceptable, indicating acceptable convergent validity. Moreover, all factor loadings were greater than 0.6, further suggesting good convergent validity. By comparing five-factor,

four-factor, three-factor, and one-factor models, we examined discriminant validity. The result in Table 2 shows that the five-factor model was the best, which ensured discriminant validity. We also examined the discriminant validity by comparing the square roots of the average variance extracted (AVE) with the correlations of constructs. As shown in Table 3, the square roots of AVE were greater than the correlation coefficients of each construct, further indicating that the discriminant validity was satisfactory.

4 | RESULTS

4.1 | Descriptive statistics and correlations

We calculated the variables' means, standard deviations (S.D.), and correlations. As presented in Table 3, search breadth is positively correlated with exploitative green innovation ($r = 0.351$, $p < .001$) and exploratory green innovation ($r = 0.249$, $p < .01$), respectively. Moreover, search depth is also positively correlated with exploitative green innovation ($r = 0.298$, $p < .001$) and exploratory green innovation ($r = 0.300$, $p < .001$). Firm scale has a significant correlation with exploitative green innovation ($r = 0.329$, $p < .001$) and exploratory green innovation

TABLE 2 Convergent validity and discriminant validity

Model	Variable	χ^2	df	RMSEA	TLI	CFI
One-factor model	SB + SD + ROC + EIGI + ERGI	2251.112	152	0.273	0.265	0.347
Three-factor model	SB + SD, ROC, EIGI + ERGI	1327.768	149	0.207	0.579	0.633
Four-factor model	SB + SD, ROC, EIGI, ERGI	850.437	146	0.161	0.743	0.781
Four-factor model	SB, SD, ROC, EIGI + ERGI	861.027	146	0.163	0.739	0.778
Five-factor model	SB, SD, ROC, EIGI, ERGI	382.246	142	0.096	0.910	0.925

TABLE 3 Means, standard deviations, and correlations

Variable	1	2	3	4	5	6	7	8	9	10	11
1. Industry 1	1.00										
2. Industry 2	-0.155*	1.00									
3. Industry 3	-0.185*	-0.267***	1.00								
4. Industry 4	-0.215**	-0.310***	-0.371***	1.00							
5. Firm age	-0.054	-0.162*	0.080	-0.011	1.00						
6. Firm scale	-0.093	-0.004	0.199**	-0.172*	0.337***	1.00					
7. SB	0.029	0.035	0.211**	-0.241**	0.045	0.232**	0.869				
8. SD	0.107	-0.002	0.083	-0.175*	-0.044	0.017	0.325***	0.866			
9. ROC	-0.109	0.049	0.252**	-0.229**	0.183*	0.192**	0.236**	0.224**	0.909		
10. EIGI	0.057	0.072	0.099	-0.282***	0.101	0.329***	0.351***	0.298***	0.270***	0.862	
11. ERGI	-0.071	-0.022	0.070	-0.151*	0.067	0.239**	0.249**	0.300***	0.476***	0.487***	0.849
Mean	0.097	0.183	0.242	0.301	3.538	3.156	3.828	3.797	3.669	3.766	3.532
S.D.	0.296	0.388	0.429	0.460	1.231	1.140	0.834	0.662	0.845	0.862	0.858

Note. The diagonal numbers in bold are the square root of average variance extracted.

* $p < .05$. ** $p < .01$. *** $p < .001$.

TABLE 4 Results for testing hypotheses

	EIGI			ERGI								
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9	Model 10	Model 11	Model 12
Constant	3.224***	1.476**	1.957***	2.220***	2.548***	2.702***	3.410***	1.681***	2.317***	3.129***	3.340***	3.671***
Control Variable												
Industry 1	0.053	-0.016	0.007	0.053	0.012	0.042	-0.452	-0.519*	-0.479*	-0.339	-0.364	-0.357
Industry 2	-0.037	-0.033	0.038	0.022	-0.007	-0.008	-0.363	-0.348	-0.243	-0.292	-0.311	-0.325*
Industry 3	-0.110	-0.178	-0.050	-0.088	-0.062	-0.130	-0.271	-0.319	-0.142	-0.262	-0.246	-0.302*
Industry 4	-0.472**	-0.360*	-0.242	-0.208	-0.249	-0.225	-0.464*	-0.356*	-0.187	-0.083	-0.109	-0.102
Firm age	0.001	0.012	0.024	0.009	0.007	-0.003	-0.024	-0.012	0.008	-0.038	-0.039	-0.050
Firm scale	0.225***	0.195***	0.185**	0.179**	0.171**	0.186***	0.165**	0.146*	0.135*	0.118*	0.113*	0.122**
Independent variable												
SB		0.204**	0.076	0.058	0.019	0.039		0.124	-0.049	-0.104	-0.130	-0.129
SD		0.266**	0.291**	0.260**	0.228**	0.158		0.331***	0.366***	0.268**	0.248**	0.158*
SB squared			-0.092*	-0.093*	-0.087*	-0.108**		-0.121**	-0.121**	-0.125**	-0.121**	-0.141***
SD squared			-0.088*	-0.094*	-0.096*	-0.122**		-0.138**	-0.138**	-0.157***	-0.159***	-0.186***
Moderator Variable												
ROC				0.122*	-0.017	0.024				0.377***	0.285***	0.263***
Interaction												
SB × ROC					0.064						0.049	
SB squared × ROC					0.148***						0.099**	
SD × ROC						0.229**						0.235***
SD squared × ROC						0.174**						0.197***
R ²	0.163	0.261	0.306	0.322	0.379	0.391	0.093	0.186	0.280	0.438	0.463	0.519
R ² change		0.097***	0.045**	0.016*	0.056***	0.069***		0.093***	0.094**	0.158***	0.025*	0.081***
F	5.822***	7.802***	7.718***	7.529***	8.064***	8.509***	3.059**	5.043***	6.805***	12.316***	11.399***	14.288***

p* < .05. *p* < .01. ****p* < .001.

($r = 0.239$, $p < .01$), indicating that there are differences in exploitative and exploratory green innovations among firms in terms of scale.

4.2 | Hypotheses testing

4.2.1 | Main effects

We applied hierarchical regression analyses to test the hypotheses. The inverted U-shaped relationship is examined by following Haans, Pieters, and He (2016). Suppose the relationship between dependent variable Y with the independent variable X and its squared is expressed as $Y = \beta_0 + \beta_1 X + \beta_2 X^2$, the inverted U-shaped relationship should meet three conditions. First, β_2 , the coefficient of squared independent variable, is negative and significant. Second, the slope of the curve must be sufficiently steep at both ends of the data range. Suppose the value of independent variable X range from X_L at the low end to X_H at the high end. An inverted U-shaped relationship is to show that the slope at X_L , which is $\beta_1 + 2\beta_2 X_L$, is positive and significant, and the slope at X_H , which is $\beta_1 + 2\beta_2 X_H$, is negative and significant. Third, the turning point of the inverted U curve, $-\beta_1/2\beta_2$, needs to be located well within the data range.

Hypotheses 1a and 1b predict that search breadth has an inverted U-shaped relationship with exploitative and exploratory green innovations. As Models 3 and 9 displayed in Table 4, search breadth squared has significantly negative effects on both exploitative green innovation ($\beta = -0.092$, $p < .05$) and exploratory green innovation ($\beta = -0.121$, $p < .01$), satisfying the first condition. We computed the turning point of the curve as 0.413 and -0.202 , respectively. After mean-centered, the range of search breadth is from -3.390 to 1.405 . When the value of search breadth is -3.390 , the slopes, which are 0.670 for exploitative green innovation and 0.771 for exploratory green innovation, are all positive and significant; and when the value of search breadth is 1.405 , the slopes, which are -0.183 for exploitative green innovation and -0.389 for exploratory green innovation, are negative and significant, indicating the second condition is satisfied. Moreover, both turning point 0.413 and -0.202 are located within the range of search breadth (-3.390 to 1.405), meeting the third condition. Hence, Hypotheses 1a and 1b are fully supported.

Models 3 and 9 in Table 4 show that search depth squared has significantly negative effects on both exploitative green innovation

($\beta = -0.088$, $p < .05$) and exploratory green innovation ($\beta = -0.138$, $p < .01$), satisfying the first condition. We computed the turning point of the curve as 1.653 and 1.326 , respectively. After mean-centered, the range of search depth is from -2.713 to 1.816 . When the value of search depth is -2.713 , the slopes, which are 0.768 for exploitative green innovation and 1.115 for exploratory green innovation, are all positive and significant; and when the value of search depth is 1.816 , the slopes, which are -0.029 for exploitative green innovation and -0.135 for exploratory green innovation, are negative and significant, indicating the second condition is satisfied. The turning point of 1.653 and 1.326 are both located within the range of search depth (-2.713 to 1.816), the third condition also satisfy. Thus, Hypotheses 2a and 2b are supported.

4.2.2 | Moderating effects

To test the moderation of resource orchestration capability in the inverted U-shaped relationships, we adapted the procedure proposed by Haans et al. (2016). Given that the curve is expressed as $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 XZ + \beta_4 X^2 Z + \beta_5 Z$, the moderation with inverted U-shaped should meet two conditions. One is that there is a significant shift in the turning point. Specifically, if $\beta_1 \beta_4 - \beta_2 \beta_3$ is positive, the turning point will move to the right as Z increases, and if $\beta_1 \beta_4 - \beta_2 \beta_3$ is negative, the turning point will move to the left. The other is that there is a significant flattening or steepening of the curve. A flattening occurs when β_4 is positive, and a steepening occurs when β_4 is negative.

Hypotheses 3a and 3b predict that resource orchestration capability moderates and flattens the relationships between search breadth and exploitative and exploratory green innovations. Models 5 and 11 in Table 4 show that resource orchestration capability significantly flattens the effect of search breadth on both exploitative green innovation ($\beta = 0.148$, $p < .001$) and exploratory green innovation ($\beta = 0.099$, $p < .01$). We computed the shift of search breadth as resource orchestration capability increases from low level to high level, the turning point move to right for exploitative green innovation (0.008), and move to left for exploratory green innovation (-0.007). Hence, Hypotheses 3a and 3b are supported. Figure 2 graphically depicts the curvilinear relationship between search breadth, exploitative, and exploratory green innovations for different levels of

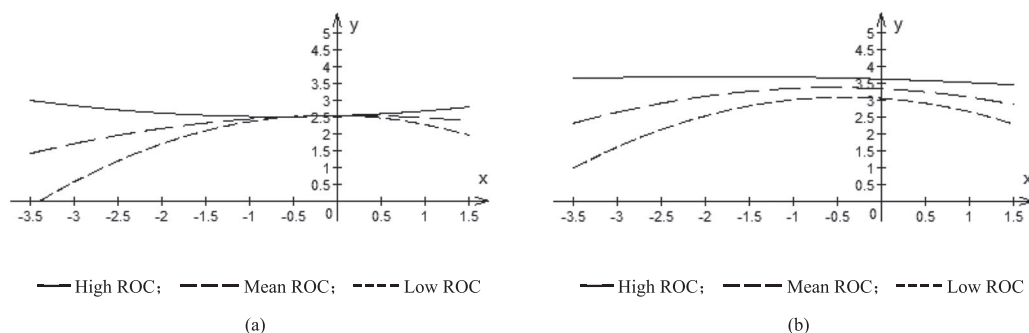


FIGURE 2 Moderation effects of resource orchestration capability between search breadth and exploitative green innovation (a), exploratory green innovation (b)

resource orchestration capability. As displayed in Figure 2a, at the higher level of resource orchestration capability, the curve of search breadth and exploitative green innovation moves to right and flattens; thus, high resource orchestration capability increases the positive effect of narrow search and reduces the negative effect of wider search on exploitative green innovation. However, at lower level of resource orchestration capability, the curve steepens, the positive effect of narrow search on exploitative green innovation is mitigated, and the negative effect of wider search is strengthened. As exhibited in Figure 2b, at the higher level of resource orchestration capability, the inverted U-shaped of search breadth and exploratory green innovation moves to left and flattens; thus, high resource orchestration capability increases the positive effect of narrow search and reduces the negative effect of wider search on exploratory green innovation. At the lower level of resource orchestration capability, the curve steepens, meaning the positive effect of narrow search is alleviated and the negative effect of wider search is strengthened. Accordingly, Hypotheses 3a and 3b are further supported, suggesting that a high level of resource orchestration capability flattens the inverted U-shaped relationships between search breadth, exploitative, and exploratory green innovations.

Further, Models 6 and 12 in Table 4 indicate that resource orchestration capability flattens the effect of search depth on both exploitative green innovation ($\beta = 0.174, p < .01$) and exploratory green innovation ($\beta = 0.197, p < .001$). We calculated the shift of search depth as resource orchestration capability increases. The turning point moves to right for both exploitative green innovation (0.055) and exploratory green innovation (0.075). Hence, Hypotheses 3c and 3d are confirmed. Figure 3 graphically illustrates the curvilinear relationships between search depth, exploitative green innovation and exploratory green innovation for different levels of resource orchestration capability. As shown in Figure 3a,b, at the higher level of resource orchestration capability, the two curves of search depth and exploitative green innovation, exploratory green innovation both move to right and flatten to almost linear, high resource orchestration capability therefore increases the positive effects and reduces the negative effects of search depth on both exploitative and exploratory green innovations. At the lower level of resource orchestration capability, the two curves steepen; hence, the positive effects of search depth on both exploitative and exploratory green innovations are mitigated,

and the negative effects are strengthened. Accordingly, Hypotheses 3c and 3d are further supported, suggesting that a high level of resource orchestration capability flattens the inverted U-shaped relationship between search depth and exploitative and exploratory green innovations.

5 | DISCUSSIONS

Drawing on resource-based theory and resource orchestration theory, we empirically investigate the relationship between boundary-spanning search and green innovation and also examine the moderator role of resource orchestration capability. We focus on two characteristics of boundary-spanning search: search breadth and search depth, and classify green innovation into exploitative green innovation and exploratory green innovation. The results indicate that boundary-spanning search has a double-edged sword effect on firms' green innovation. Specifically, search breadth has an inverted U-shaped effect on both exploitative green innovation and exploratory green innovation. Furthermore, search depth also displays an inverted U-shaped impact on exploitative green innovation and exploratory green innovation. Our results confirm the viewpoint that resource orchestration capability plays a vital moderating role between boundary-spanning search and green innovation. On the one hand, resource orchestration capability moderates the links between search breadth and exploitative and exploratory green innovations and flattens the inverted U-shaped relationship when resource orchestration capability is higher. On the other hand, resource orchestration capability plays a moderating role in the relationships between search depth and exploitative and exploratory green innovations, and the relationships become almost linear and no longer inverted U-shaped under the high level of resource orchestration capability.

5.1 | Theoretical implications

First, we find that two characteristics of boundary-spanning search display inverted U-shaped impact on both exploitative and exploratory green innovations. Our results are consistent with Laursen and Salter (2006), Zhang and Hu (2017), and Ardito and Petruzzelli (2017), who suggest that search breadth and search depth have inverted U-shaped

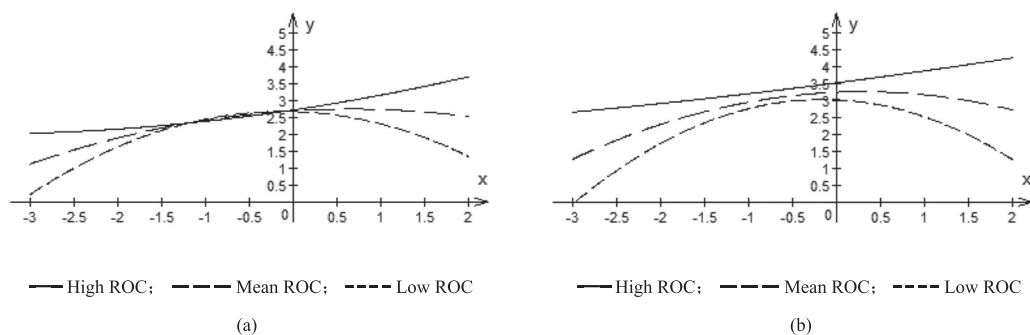


FIGURE 3 Moderation effects of resource orchestration capability between search depth and exploitative green innovation (a), exploratory green innovation (b)

effects on innovation. We extend this curvilinear relationship into the environmental sustainability context and classify green innovation into exploitative green innovation and exploratory green innovation. Existing studies have shown that there are contradictory results between the two characteristics of boundary-spanning search and innovation. Based on resource-based theory, our study indicates that too little and too much boundary-spanning search are not beneficial for firms to implement green innovation, and the relationship between boundary-spanning search and green innovation is not linear. The findings provide explanations to the conflicts that some researches propose the effect of boundary-spanning search on innovation is positive, whereas others suggest the relationship is negative. The result further provides possible evidence that search breadth and search depth have double-edged sword effects on exploitative and exploratory green innovations and suggests that how much boundary-spanning search is suitable and optimal for firms' green innovation.

Second, our study emphasizes the importance of resource orchestration capability. We prove that resource orchestration capability moderates and flattens the inverted U-shaped links between search breadth and search depth with exploitative and exploratory green innovations. Specifically, from the resource orchestration perspective, our result shows that the influences of the two search characteristics on green innovation are different. With a high level of resource orchestration capability, the relationships of search breadth with exploitative and exploratory green innovations are flattened and still inverted U-shaped, but the relationships between search depth with exploitative and exploratory green innovations are almost linear and no longer inverted U-shaped. The result is similar with the results of Chiang and Hung (2010), Garriga, Von Krogh, and Spaeth (2013), and Flor et al. (2018), who think that firms' capability to integrate, combine, and reconfigure external and internal knowledge resources is very important, which helps firms effectively reap the advantage of external knowledge breadth and depth to conduct radical innovation. This result implies that resource orchestration capability is an important organizational capability that orchestrates knowledge to facilitate exploitative and exploratory green innovations, which can help firms to efficiently utilize the knowledge acquired from boundary search. This result enriches the current research and provides a more comprehensive understanding of resource orchestration capability, and uncovers the mechanism through which boundary-spanning search and resource orchestration capability interact with each other to shape green innovation. To our knowledge, there is no research examining resource orchestration capability in the context of boundary-spanning search and green innovation. This study makes an important contribution by filling this void.

5.2 | Managerial implications

Our research provides two managerial implications. First, our results suggest boundary-spanning search has inverted U-shaped relationships with exploitative and exploratory green innovations, and too much or too little knowledge search will be harmful to green

innovation. Hence, in order to facilitate green innovation, firms should not only emphasize acquiring external knowledge by cooperating with external stakeholders but also pay attention to the degree of search breadth and search depth. Moreover, firms should balance the benefits and costs of knowledge searching from different organization and industries and keep external search at a suitable level, such that the search can be cost-beneficial for facilitating green innovation.

Second, our study reveals that resource orchestration capability does not directly impact on green innovation, but as a moderator indirectly influence green innovation by interacting with boundary-spanning search. This highlights the importance of resource orchestration capability in boundary-spanning search and green innovation. To accelerate exploitative and exploratory green innovations, firms should take resource orchestration capability into consideration and should enhance their resource orchestration capability. For example, firms can create formal and informal processes to facilitate knowledge transfer and take measures to encourage organizational learning, in such a way that resource orchestration capability can be promoted.

5.3 | Limitations and future research

There are several limitations in our research. First, our survey is based on the self-report measures, which may involve common method bias. Although we took several remedies to reduce this bias, the problem may still exist (Hair, Black, Babin, Anderson, & Tatham, 2006). Hence, we suggest that the future study should collect data from multiple sources. Second, owing to our sample in the manufacturing industry, the findings of this study may be industry specific. Future research needs to confirm the relationships in other industries. Third, our research only investigates the moderating effect of resource orchestration capability among many possible moderators. Future research can test other factors, such as green absorptive capability, stakeholders' pressure, and green innovation intent, and examine how those boundary conditions impact on the links between boundary-spanning search and green innovation.

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